

DAVID (YOON SUK) KANG

CONTACT INFORMATION	Building S4-1, Room 314 Chungdae-ro 1, Seowon-gu, Cheongju-si, Chungcheongbuk-do, 28644, Korea	Tel: +82-43-261-2237 Email: dyskang@cbnu.ac.kr Homepage: https://dyskang.github.io
RESEARCH INTERESTS	My primary research interests lie in data mining applied to diverse graph data types (<i>e.g.</i> , <i>conventional</i> , <i>signed</i> , <i>knowledge</i> , and <i>hypergraph</i> structures), with a particular emphasis on uncovering knowledge from real-world networks. • Conventional Graph: Community detection (CIKM'20, KBS'22) • Signed Graph: Community detection (ICDM'21, TKDE'23); Representation learning (TKDD'24) • Hypergraph: Representation learning (ICDM'25°, DMKD'25°); Hypergraph analysis (WWW'24) (°: <i>Under Review</i>)	
EDUCATION	Hanyang University , Seoul, Korea • <i>Ph.D. in Computer Science</i> – Thesis: Graph Reinforcement for Accurate Community Detection and Embedding on Graphs and Hypergraphs – Advisor: Prof. Sang-Wook Kim	Mar. 2013 – Feb. 2022
	Hanyang University , Seoul, Korea • <i>B.S. in Computer Science</i>	Mar. 2007 – Feb. 2013
POSITIONS	Chungbuk National University , Cheongju-si, Chungcheongbuk-do, Korea • <i>Assistant Professor, School of Computer Science</i>	Sep. 2024 – Present
RESEARCH EXPERIENCES	University of Michigan , Ann Arbor, MI, USA • <i>Postdoctoral Researcher, School of Information</i> – Topic: Data Mining on Large-Scale Hypergraph – Advisor: Prof. Qiaozhu Mei	May 2022 – Aug. 2024
	The Pennsylvania State University , University Park, PA, USA • <i>Visiting Scholar, College of Information Sciences and Technology</i> – Topic: Improving the Accuracy of Community Detection – Advisor: Prof. Dongwon Lee	Oct. 2019 – Feb. 2020
AWARDS & HONORS	Received the Best Paper Award in Samsung Research Project • Samsung Electronics Co., Ltd.	2022
	Received the Outstanding Ph.D. Dissertation Award • Research Institute of Industrial Science, Hanyang University	2022
	Awarded the NAVER Ph.D. Fellowship • Naver Corporation	2021
	Received the Best Paper Award • KIPS Spring Conference	2021
	Received the ACM SIGIR Student Travel Award • ACM International Conference on Information and Knowledge Management (ACM CIKM)	2017, 2020
	Received the ACM SIGAPP Student Travel Award • ACM Symposium on Applied Computing (ACM SAC)	2016
	Awarded the NHN&HYU Ph.D. Fellowship • NHN Corporation	2015
	Received the Best Paper Award • IEEE International Conference on Network Infrastructure and Digital Content (IEEE IC-NIDC)	2014

Preprinted/Submitted Papers (* indicates equal contributions)

- [1] (Tentative title) Learning on Hypergraphs
So-Bin Jung*, David Yoon Suk Kang*, and Sang-Wook Kim
Under Review at One of Top-Tier DM Conferences
- [1] Revisiting Clique and Star Expansions in Hypergraph Representation Learning: Observations, Problems, and Solutions
David Yoon Suk Kang, Eujeanne Kim, Kyungsik Han, and Sang-Wook Kim
Under Review at Data Mining and Knowledge Discovery

International Conference and Journal Papers

- [13] Trustworthiness-Driven Graph Convolutional Networks for Signed Network Embedding
Min-Jeong Kim*, Yeon-Chang Lee*, David Y. Kang, and Sang-Wook Kim
ACM Transactions on Knowledge Discovery from Data (SCIE Journal, 2024)
- [12] Low Mileage, High Fidelity: Evaluating Hypergraph Expansion Methods by Quantifying the Information Loss
David Y. Kang, Qiaozhu Mei, and Sang-Wook Kim
WWW 2024 (*ACM Web Conference*)
Full Paper (Acceptance Rate $\approx 20\%$)
Selected for Oral Presentation
- [11] A Framework for Accurate Community Detection on Signed Networks Using Adversarial Learning
David Y. Kang, Woncheol Lee, Yeon-Chang Lee, Kyungsik Han, and Sang-Wook Kim
IEEE Transactions on Knowledge and Data Engineering (Top 5% SCIE Journal, 2023)
- [10] Community Reinforcement: An Effective and Efficient Preprocessing Method for Accurate Community Detection
Yoonsuk Kang, Jun-Seok Lee, Won-Yong Shin, and Sang-Wook Kim
Knowledge-Based Systems (Top 10% SCIE Journal, 2022)
- [9] Adversarial Learning of Balanced Triangles for Accurate Community Detection on Signed Networks
Yoonsuk Kang*, Woncheol Lee*, Yeon-Chang Lee, Kyungsik Han, and Sang-Wook Kim
ICDM 2021 (*IEEE International Conference on Data Mining*)
Short Paper (Acceptance Rate $\approx 20\%$)
- [8] FORESEE: An Effective and Efficient Framework for Estimating the Execution Times of IO Traces on the SSD
Yoonsuk Kang, Yong-Yeon Jo, Jaehyuk Cha, Wan D. Bae, Wonjun Lee, and Sang-Wook Kim
IEEE Transactions on Computers (SCIE Journal, 2021)
- [7] CR-Graph: Community Reinforcement for Accurate Community Detection
Yoonsuk Kang, Jun-Seok Lee, Won-Yong Shin, and Sang-Wook Kim
CIKM 2020 (*ACM International Conference on Information and Knowledge Management*)
Short Paper (Acceptance Rate $\approx 25\%$)
- [6] A Framework for Estimating Execution Times of IO Traces on SSDs
Yoonsuk Kang, Yong-Yeon Jo, Jaehyuk Cha, Wan. D. Bae, and Sang-Wook Kim
CIKM 2017 (*ACM International Conference on Information and Knowledge Management*)
Short Paper (Acceptance Rate $\approx 28\%$)
- [5] The uFLIP Benchmark Revisited for Evaluating SSDs
Yoonsuk Kang, Yong-Yeon Jo, Jaehyuk Cha, Sang-Wook Kim, and Young Kyun Shin
International Journal of Communication Systems (SCIE Journal, 2016)
- [4] A Methodology for Estimating Execution Times of IO Traces in SSDs
Yoonsuk Kang
SAC 2016 (*ACM Symposium on Applied Computing*)

- [3] Exploiting the uFLIP Benchmark for Analyzing SSDs Performance
Yoonsuk Kang, Yong-Yeon Jo, Jaehyuk Cha, Sang-Wook Kim, and Young Kyun Shin
IC-NIDC 2014 (*IEEE International Conference on Network Infrastructure and Digital Content*)
Received the Best Paper Award
- [2] Running Data Mining Algorithms on SSDs
Yoonsuk Kang, Yong-Yeon Jo, Duck-Ho Bae, and Sang-Wook Kim
EDB 2013 (*International Conference on Emerging Databases-Technologies, Applications, and Theory*)
- [1] Selecting Similar Users in Collaborative Filtering
 Sang-Chul Lee, Yoonsuk Kang, Seihyun Jeong, Min-Hee Jang, Young-Sup Hwang, and Sang-Wook Kim
ICGHIT 2013 (*International Conference on Green and Human Information Technology*)

Domestic Conference and Journal Papers

- [13] Constructing a Graph-Based arXiv Dataset By Reflecting the Research Trend in Computer Science
 Juhyun Jeon, David Y. Kang, and Sang-Wook Kim
ASK 2024
- [12] Evaluating the Performance of Hypergraph Embedding Methods According to Hypergraph Sparsity
 So-Bin Jung, David Y. Kang, and Sang-Wook Kim
ASK 2024
- [11] CoAID+: COVID-19 News Cascade Dataset for Social Context Based Fake News Detection
 Soeun Han, Yoonsuk Kang, Yunyong Ko, Jiwon Ahn, Yusim Kim, Seongsu Oh, Heejin Park,
 and Sang-Wook Kim
KIPS Transactions on Software and Data Engineering (KCI Journal, 2022)
- [10] COVID-19 Cascade Dataset for Fake News Detection
 Soeun Han, Yoonsuk Kang, Yunyong Ko, Jiwon Ahn, Yusim Kim, Seongsu Oh, Heejin Park,
 and Sang-Wook Kim
KIPS Spring Conference 2021
Received the Best Paper Award
- [9] A Preprocessing Method for Accurate Link Prediction on Social Networks
 Seungbeom Son, Yeonsuk Choi, Yoonsuk Kang, and Sang-Wook Kim
KIPS Fall Conference 2020
- [8] Performance Comparison of Similarity-Based Link Prediction in Social Networks
 Jun-Seok Lee, Yoonsuk Kang, and Sang-Wook Kim
KCC 2019
- [7] Performance Comparison of Community Detection Algorithms in Social Networks
 Jun-Seok Lee, Yoonsuk Kang, and Sang-Wook Kim
KCC 2018
- [6] A Method for Analyzing Features that Affect the Performance of SSD
Yoonsuk Kang, Yong-Yeon Jo, and Sang-Wook Kim
KIPS Spring Conference 2018
- [5] Community Detection by Sub-Community and CScan
 Chunghyeon Cho, Gunjoo Ahn, Yoonsuk Kang, Jiwon Hong, and Sang-Wook Kim
KDBC 2018
- [4] A Methodology for Estimating Execution Times of IO Traces on SSDs
Yoonsuk Kang, Yong-Yeon Jo, Jaehyuk Cha, Wan D. Bae, and Sang-Wook Kim
KCC 2017

- [3] Analyzing the Performance of SSDs in OLTP Environment
Seoung-Hun Jeong, Jae-Sung Lee, Yoonsuk Kang, Yong-Yeon Jo, Duck-Ho Bae, Sang-Wook Kim, Juyoung Kang, and Jahyuk Cha
KIISE Fall Conference 2013
- [2] Analysis on I/O Trace Replayer for SSD Performance Evaluation
Inhyuk Yee, Kyuhwan Lee, Yoonsuk Kang, Yong-Yeon Jo, and Sang-Wook Kim
KIPS Fall Conference 2013
- [1] A Method for Selecting Similar Users for Collaborative Filtering
Yoonsuk Kang, Seihyun Jeong, Sang-Chul Lee, Min-Hee Jang, Sang-Wook Kim
KIPS Fall Conference 2012

INVITED TALKS

Self-Supervised Hypergraph Representation Learning

- Invited Talk @Hanyang Univ., June. 2025

Evaluating Hypergraph Expansion Methods by Quantifying the Information Loss

- Invited Talk @Chung-Ang Univ., Nov. 2024

Knowledge Discovery from Real-world Relationships

- Invited Talk @Hanyang Univ., Aug. 2024

Adversarial Learning of Balanced Triangles for Accurate Community Detection on Signed Networks

- Invited Talk @ METU-HYU Joint Workshop, Dec. 2022

FORESEE: An Effective and Efficient Framework for Estimating the Execution Times of IO Traces on SSDs

- Invited Talk @ Waseda-UMS-Hanyang-UKM (WUHU) Joint Workshop, Dec. 2017

TEACHING

Undergraduate School

- Information Retrieval (5119006) Fall 2025
- Machine Learning (5119008) Spring 2025
- Data Structure (5118006) Spring 2025
- Data Science (5119010) Spring 2025
- Server Programming (5120017) Fall 2024 – 2025
- Fundamental Computer Programming (0914002) Fall 2024 – 2025
- Algorithms (5118013) Fall 2024 – 2025

PROFESSIONAL SERVICES

Program Committee Member

- ACM International Conference on Information and Knowledge Management (**CIKM**) 2025
- ACM Symposium on Applied Computing (**SAC**) 2023 – 2025

Conference Reviewer

- ACM Conference on Research and Development in Information Retrieval (**SIGIR**) 2024
- ACM Web Conference (**WWW**) 2023, 2024
- ACM Conference on Knowledge Discovery and Data Mining (**KDD**) 2021 – 2024
- IEEE International Conference on Data Mining (**ICDM**) 2022 – 2024
- ACM International Conference on Information and Knowledge Management (**CIKM**) 2019, 2020, 2025
- ACM Symposium on Applied Computing (**SAC**) 2023, 2024
- International AAAI Conference on Web and Social Media (**ICWSM**) 2017

Journal Reviewer

- Journal of Supercomputing 2023

PATENTS

Granted Patents

- Adversarial Learning of Balanced Triangles for Accurate Community Detection on Signed Networks
Registration Number: **KR10-2764877** Feb. 2025

- Method for Reconfiguration of a Community in a Network Including a Plurality of Networks and an Electronic Device for the Method
Registration Number: **KR10-2409160** Jun. 2022
- A Feature Extraction Apparatus and Method for Predicting the Execution time of the Query Input and Output Trace
Registration Number: **KR10-2249832** May 2021
- A SSD Performance Evaluation Apparatus and Method for Predicting the Execution time of the Query Input and Output Trace
Registration Number: **KR10-1950801** Feb. 2019
- Method for Selecting Similar Users for Collaborative Filtering Based on Earth Mover's Distance
Registration Number: **KR10-1620659** May 2016

Filed Patents

- Method and System for Measuring the Amount of Information Loss of a Graph Obtained Through a Hypergraph Expansion Method
Application Number: **KR10-2023-0109155** Aug. 2023
- Hypergraph Embedding Method and Systems Based on Graph Convolutional Networks Considering Relationships of Multiple-users
Application Number: **KR10-2023-0065477** May 2023

REFERENCES

- Qiaozhu Mei**, *Professor* (Postdoc. Advisor) qmei@umich.edu
School of Information, University of Michigan
- Kyungsik Han**, *Associate Professor* (Collaborator) kyungsikhan@hanyang.ac.kr
Department of Data Science, Hanyang University
- Dongwon Lee**, *Professor* (Visiting Scholar Advisor) dongwon@psu.edu
College of Information Sciences and Technology, The Pennsylvania State University
- Wan D. Bae**, *Professor* (Collaborator) baew@seattleu.edu
Department of Computer Science, Seattle University
- Sang-Wook Kim**, *Professor* (Ph.D. Advisor) wook@hanyang.ac.kr
Department of Computer Science, Hanyang University